Phishing Website Detection Using Machine Learning

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1. Executive Summary

Phishing websites are fraudulent online pages designed to steal user credentials, financial data, and personal information. As digital transactions increase, phishing attacks pose serious risks to individuals and organizations. This project aims to develop an intelligent machine learning system that detects phishing websites using URL structure, domain properties, and webpage behavior features.

The dataset contains 11,430 websites with 89 features describing URL characteristics, domain information, and webpage security signals. The goal is to classify websites as **phishing** or **legitimate**, helping organizations prevent cyber fraud, protect user trust, and strengthen cybersecurity systems.

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2. Business Problem Understanding

2.1 Problem Statement

Cyber attackers create fake websites that closely resemble legitimate platforms to trick users into revealing sensitive information. Traditional blacklist methods cannot detect newly created phishing websites. A predictive detection system is needed to identify phishing websites automatically before users interact with them.

2.2 Scope

- Real-time phishing detection
- Browser/email security integration
- Fraud prevention systems
- Automated cybersecurity monitoring

2.3 Importance of Detecting Phishing Websites

- Prevents financial loss and fraud
- Protects user credentials and privacy
- Reduces brand reputation damage
- Strengthens enterprise security infrastructure

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3. Literature Review Insights

Common Phishing Characteristics

- Long and complex URLs
- Use of IP address instead of domain
- Excessive special characters
- Multiple subdomains
- Suspicious redirects

- Fake HTTPS indicators
- Short domain age
- External scripts and media links

Detection Challenges

- Phishing domains have short lifespans
- URL obfuscation techniques
- Visual similarity to legitimate sites
- Evolving attack patterns

Potential Solutions

- URL-based machine learning models
- Domain registration analysis
- Webpage behavior analysis
- Hybrid ML + blacklist systems

Dataset Exploration Report

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4. Dataset Overview

Attribute	Value
Total Records	11,430 websites
Total Features	89 columns
Feature Types	74 Integer, 13 Float, 2 Object
Target Variable	`status` (phishing / legitimate)

Target Variable Distribution

Class	Count
Phishing	5,715
Legitimate	5,715

✔ Dataset is **perfectly balanced**.

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5. Feature Categories

Feature Description and Relevance to Phishing Detection

Feature	Description	Relevance to Phishing Detection

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length_url	Total number of characters in the URL	Phishing URLs are often very long to hide suspicious parts
nb_dots	Number of dots in URL	Many dots indicate multiple subdomains, common in phishing
nb_hyphens	Count of '-' symbols in URL	Attackers use hyphens to mimic real domains
nb_at	Presence of '@' symbol	'@' redirects browser and hides actual domain
nb_slash	Number of '/' in URL	Excessive slashes suggest deep fake directory paths
ratio_digits_url	Ratio of digits in URL	Phishing URLs often contain random numbers
domain_age	Age of the domain (in days)	Phishing domains are newly registered
domain_registration_length	Duration of domain registration	Legitimate domains are registered for longer periods
whois_registered_domain	WHOIS registration status	Unregistered or hidden WHOIS info is suspicious
dns_record	Availability of DNS record	Missing DNS record suggests malicious intent
google_index	Whether the site is indexed by Google	Phishing sites are often not indexed
page_rank	Website ranking score	Low-ranked domains are usually untrustworthy
https_token	Presence of "https" token in domain	Fake domains misuse https token to appear secure
ssl_final_state	SSL certificate trust status	Invalid SSL often indicates phishing
nb_redirection	Number of redirections	Phishing pages use multiple redirects to hide origin
nb_external_redirection	Redirections to external domains	Used to send users to malicious pages
ratio_extHyperlinks	Ratio of external hyperlinks	Phishing pages link to external malicious sources
ratio_extMedia	Ratio of external media objects	External media is common in phishing pages
login_form	Presence of login form	Used to steal user credentials

Feature	Description	Relevance to Phishing Detection
popup_window	Presence of pop-up windows	Phishing uses popups to request sensitive info
iframe	Use of invisible iframe	Used to load malicious content secretly
submit_email	Email submission form	Phishing collects credentials via email forms
external_favicon	Favicon loaded from external source	Indicates fake website branding
links_in_tags	Hyperlinks inside HTML tags	Used to hide malicious redirections
ratio_intHyperlinks	Ratio of internal hyperlinks	Phishing sites often have fewer internal links
length_hostname	Length of hostname	Long hostnames indicate suspicious domains
nb_www	Count of 'www' in URL	Multiple 'www' used to confuse users
nb_com	Count of 'com' in URL	Fake domains often repeat 'com'
shortest_word_host	Shortest word length in hostname	Random short words common in phishing domains
longest_word_path	Longest word in URL path	Long random paths are suspicious
avg_word_path	Average word length in path	Irregular path words indicate phishing
char_repeat	Repeated characters in URL	Used to mimic legitimate sites
ratio_digits_host	Ratio of digits in hostname	Numeric hostnames often malicious
nb_subdomains	Number of subdomains	Many subdomains used to appear legitimate
tld_length	Length of top-level domain	Unusual TLD lengths linked with phishing
domain_in_title	Domain present in page title	Mismatch indicates phishing
domain_with_copyright	Domain in copyright text	Absence suggests fake website
nb_hyperlinks	Total hyperlinks on page	Phishing pages have fewer meaningful links
nb_images	Number of images	Low image count suggests fake site
nb_css	Number of CSS files	Minimal styling in phishing pages
nb_js	Number of JavaScript files	Excess scripts used for malicious actions
nb_self_ref	Self-referencing links	Legit sites link internally more
nb_empty_links	Empty or broken links	Poor site quality often phishing
nb_internal_links	Internal hyperlinks count	Legit sites have structured internal links
nb_external_links	External links count	Phishing uses more external redirects

Feature	Description	Relevance to Phishing Detection
status	Target variable (phishing/legitimate)	Classification output

URL Structure Features

Examples: `length_url`, `nb_dots`, `nb_hyphens`, `nb_at`, `nb_slash`, `ratio_digits_url`
➡ Long URLs and excessive symbols often indicate phishing.

Domain-Based Features

Examples: `domain_age`, `domain_registration_length`, `whois_registered_domain`, `dns_record`, `page_rank`
➡ Phishing domains are usually newly registered and poorly ranked.

Security Indicators

Examples: `https_token`, `ssl_final_state`, `google_index`
➡ Fake HTTPS tokens or unindexed domains suggest risk.

Webpage Behavior Features

Examples: `popup_window`, `iframe`, `login_form`, `external_favicon`, `submit_email`
➡ Phishing pages often request credentials via pop-ups and forms.

Link & Redirection Features

Examples: `nb_redirection`, `nb_external_redirection`, `ratio_extHyperlinks`
➡ Excess redirection is a common phishing tactic.

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6. Exploratory Data Analysis (EDA)

Key Observations

- Phishing URLs tend to have **higher URL length**.
- Higher number of special characters correlates with phishing.
- Domains with **low age** are more likely phishing.
- Websites with **many external hyperlinks** are often malicious.
- Use of IP address in URL is strongly linked to phishing.
- Redirect counts are higher in phishing samples.

Correlation Patterns

- URL length and digit ratio show positive correlation with phishing.
- Domain age negatively correlates with phishing.
- External media and redirection features show strong importance.

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7. Data Quality Assessment

Aspect	Status
Missing Values	None detected
Duplicate Records	Not significant
Data Imbalance	None (balanced dataset)
Outliers	Present in URL length and traffic features

Handling Strategy

- Outliers handled carefully using scaling.
- Feature selection applied to reduce redundancy.

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8. Key Insights from Data

- URL-based features alone carry strong predictive power.
- Domain age and registration length are highly important.
- Webpage behavioral features improve detection accuracy.
- Multi-feature analysis outperforms blacklist systems.

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9. Business Value of This Project

- Early detection of phishing threats
- Integration into browsers and firewalls
- Protection against identity theft
- Reduced financial fraud
- Enhanced customer trust and cybersecurity posture

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10. Next Phase

- Model development and comparison
- Hyperparameter tuning
- Feature importance analysis
- Deployment as API or browser plugin

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11. Tools & Technologies

Category	Tools
Language	Python
Libraries	Pandas, NumPy, Scikit-learn
Visualization	Matplotlib, Seaborn
ML Models (Planned)	Logistic Regression, Random Forest, XGBoost

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12. Author

Name: **Shravan Shidruk (#37015)**
GitHub : **https://github.com/shravanshidruk16/ITVedant**
Course: **Code B - Data Science Integrated Internship**
Week : **Week 1: Understanding the Problem and Dataset**
Institution: **ITVedant**